

Testing

kotlinlang.org/education



Testing

The software development experience we've accumulated over the many years unfortunately tells us that

- Every program contains bugs.
- If a program does not contain bugs, the algorithm that it implements contains them.
- If neither the program nor the algorithm contains bugs, no one needs the program (almost always).

Testing

However, everything is relative:

- A bug in a website might not really hurt anybody
- A bug in rocket launch calculations can result in terrible consequences and expenses ([Ariane 5](#))
- A bug in a radiation-treatment device can lead to deaths ([Therac-25](#))



Ariane 5

Testing: history

- 60s: Do exhaustive testing.
- Early 70s: Show that the program works correctly.
- Late 70s: Show that the program does not work correctly.
- 80s: Prevent defects throughout development.

Testing: goals

- Correct behavior of the product in all conditions.
- Compliance with the requirements.
- Information about the current state of the product.
- Error prevention and detection.
- Development cost reduction.

Testing: principles

- Testing demonstrates the presence of defects, but it does not prove their absence.
- The earlier the better.
- The absence of bugs is not an absolute goal.
- Many more.

Testing: types

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Others, e.g. security, compliance, etc.

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- System testing – Checking the entire system.
- Acceptance testing – Verifying system compliance with all the client requirements.

Unit testing

For each non-trivial function, their own tests are written that check that the method works correctly:

- Frequent launch expected $\Rightarrow$ should run fast
- One test $\Rightarrow$ one use case

Unit testing in Kotlin

The [JUnit5](#) framework is the most popular way to test Java and Kotlin programs.

```
dependencies {  
    ...  
    testImplementation(platform("org.junit:junit-bom:5.8.2"))  
    testImplementation("org.junit.jupiter:junit-jupiter:5.8.2")  
}  
  
tasks.test {  
    useJUnitPlatform()  
}
```

To run tests: `./gradlew test`

Unit testing in Kotlin

```
class MyTests {  
    @Test  
    @DisplayName("Check if the calculator works correctly")  
    fun testCalculator() {  
        Assertions.assertEquals(  
            3,  
            myCalculator(1, 2, "+"),  
            "Assertion error message"  
        )  
    }  
}
```

Unit testing in Kotlin

```
class MyParametrizedTests {  
    companion object {  
        @JvmStatic  
        fun calculatorInputs() = listOf(  
            Arguments.of(1, 2, "+", 3),  
            Arguments.of(0, 5, "+", 5),  
        )  
    }  
    @ParameterizedTest  
    @MethodSource("calculatorInputs")  
    fun testCalculator(a: Int, b: Int, op: String, expected: Int) {  
        Assertions.assertEquals(expected, myCalculator(a, b, op), "Assertion error message")  
    }  
}
```

Unit testing in Kotlin

There are many annotations for tests customization.

- `@BeforeEach` – Methods with this annotation are run before each test.
- `@AfterEach` – Methods with this annotation are run after each test.
- `@BeforeAll` – Methods with this annotation are run before all tests in the class.
- `@AfterAll` – Methods with this annotation are run after all tests in the class.

Code quality

Often, testing includes checking not only the correctness of the functionality but also the **quality** of the code itself.

Static code analyzers such as [detekt](#), [ktlint](#), and [diktat](#) exist to help you avoid having to do this manually.

The build of static analyzers should also be green.

Thank you

